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SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE: MLA0405
Slot :D

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I M.Ajitha(192525391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:
M.Ajitha



Scenario-Based Questions CO-AT2

QUESTIONS: 05

Total Marks:25

1.Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer:

Unsupervised Training of Neural Networks – Autoencoders

Recommended Deep Learning Technique: Autoencoder (Unsupervised Learning)

For this problem, the best deep learning technique is an Autoencoder because the hospital has thousands of ECG records without labels (no indication of whether they are normal or abnormal). Autoencoders learn the hidden patterns of normal ECG signals without requiring labeled data. After training, they reconstruct the input ECG. If an ECG is abnormal, the reconstruction error becomes high, making it easy to detect unusual heart patterns.

Justification:

- Autoencoders work with unlabeled data, making them suitable for this problem.
- They automatically learn the important features of ECG signals.
- The model compresses the ECG into a smaller representation and then reconstructs it.
- Normal ECGs are reconstructed accurately because the model has learned their patterns.
- Abnormal ECGs produce a high reconstruction error, which helps identify anomalies.
- This method reduces the need for manual labeling by doctors.
- It is widely used in healthcare for anomaly detection and medical signal analysis.

Workflow:

Step 1: Collect Data

- Gather thousands of unlabeled ECG records from hospital patients.
- Most ECGs are expected to represent normal heart activity.

Step 2: Data Preprocessing

- Remove unwanted noise from ECG signals.
- Normalize the ECG values.
- Convert all ECG records to the same length and format.

Step 3: Train the Autoencoder

- The Encoder compresses the ECG into important features.
- The compressed representation is called the latent (hidden) representation.

- The Decoder reconstructs the ECG from these features.
- During training, the model learns the structure of normal ECG signals.

Step 4: Test New ECG Records

- A new ECG is passed through the trained Autoencoder.
- The reconstructed ECG is compared with the original ECG.

Step 5: Calculate Reconstruction Error

- If the reconstruction error is low, the ECG is considered Normal.
- If the reconstruction error is high, the ECG is considered Abnormal.

Step 6: Final Decision

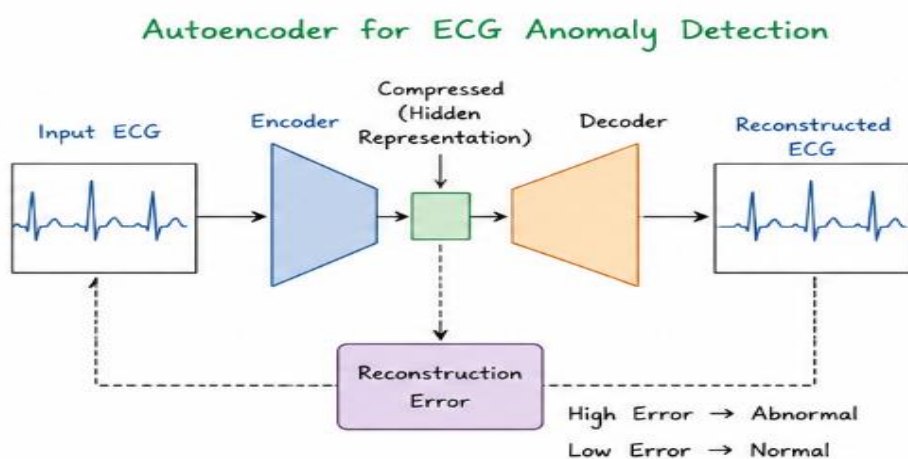
- Normal ECGs are stored in the database.
- Abnormal ECGs are sent to cardiologists for further diagnosis and treatment.

Advantages of Using Autoencoders

- Works without labeled training data.
- Learns meaningful features automatically.
- Detects abnormal ECG patterns effectively.
- Reduces manual effort for doctors.
- Improves the speed and accuracy of medical diagnosis.
- Suitable for large healthcare datasets.

Conclusion

Autoencoders are the most suitable deep learning technique for detecting abnormal ECG patterns from thousands of unlabeled records. They learn the characteristics of normal ECG signals and identify abnormalities based on reconstruction error. This approach provides an efficient, accurate, and cost-effective solution for automatic ECG anomaly detection in hospitals.



2.Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer:

Introduction:

Retail companies collect a large amount of customer purchase data, such as products purchased, purchase frequency, amount spent, payment method, discounts used, and shopping preferences. This data contains hundreds or even thousands of features, making it high-dimensional and difficult to analyze.

An Autoencoder, a deep learning model used in unsupervised learning, can reduce the number of features while preserving the most important information. The reduced data (latent representation) is then used for customer segmentation, where customers with similar buying behavior are grouped together.

Why Autoencoder is Suitable?

An Autoencoder is recommended because:

- It works with unlabeled customer data.
- It automatically learns useful patterns from purchase history.
- It compresses high-dimensional data into fewer important features.
- It removes unnecessary and redundant information.
- It improves the performance of clustering algorithms like K-Means.
- It helps businesses understand customer behavior more effectively.

Working of Autoencoder

Step 1: Data Collection

- Collect customer purchase records.
- Each customer has many features such as:
 - Products purchased
 - Purchase frequency
 - Amount spent
 - Brand preference
 - Payment method
 - Shopping category

Step 2: Data Preprocessing

Before training the model:

- Remove missing values.
- Normalize the data.
- Convert categorical values into numerical form.
- Prepare the dataset for training.

Step 3: Encoder

The Encoder compresses the original customer purchase data into a smaller set of important features.

Example:

Original Features (1000 Features)

↓

Compressed Features (30 Features)

The encoder keeps only the most meaningful information.

Step 4: Latent Representation

The compressed data is called the latent representation (or bottleneck layer).

It contains:

- Customer purchasing habits
- Spending behavior
- Product preferences
- Buying patterns

This representation is much smaller but still preserves useful information.

Step 5: Decoder

The Decoder reconstructs the original purchase data from the compressed representation.

The Autoencoder learns by minimizing the difference between:

- Original purchase data
- Reconstructed purchase data

This difference is called the Reconstruction Error.

Step 6: Dimensionality Reduction

After training, only the Encoder is used.

Instead of using all 1000 features, only the compressed features are taken.

Example:

1000 Features

↓

30 Important Features

This makes the data easier to analyze.

Step 7: Customer Segmentation

The reduced features are given to clustering algorithms such as K-Means.

Customers are grouped into segments like:

- Premium Customers
- Regular Customers
- Discount Shoppers
- Occasional Buyers
- High-Value Customers

The company can then design personalized marketing strategies for each group.

Advantages of Autoencoder:

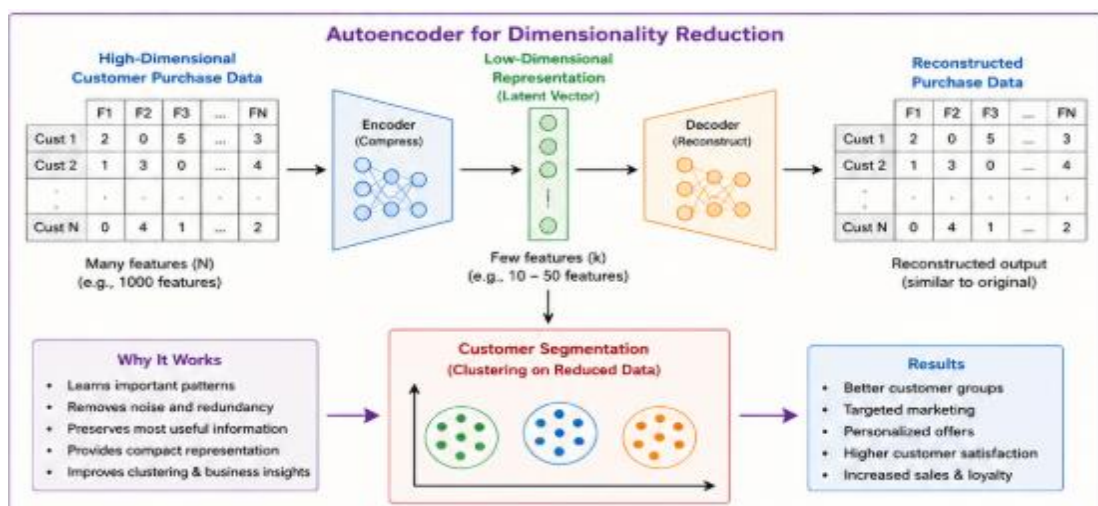
- Performs dimensionality reduction automatically.
- Learns hidden customer purchasing patterns.
- Removes redundant and noisy features.
- Improves clustering accuracy.
- Works without labeled data.
- Reduces storage and computation time.
- Helps in personalized marketing.
- Improves customer relationship management.

Applications:

- Customer segmentation
- Product recommendation systems
- Personalized advertisements
- Market basket analysis
- Customer behavior prediction
- Sales forecasting

Conclusion:

Autoencoders are an effective deep learning technique for representation learning and dimensionality reduction. They compress high-dimensional customer purchase data into a smaller and more meaningful representation while preserving important information. The reduced data helps clustering algorithms create accurate customer segments, enabling retailers to improve marketing strategies, customer satisfaction, and business performance.



3.Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer:

Introduction:

Facial recognition is one of the most important applications of Deep Learning. A facial image contains complex features such as edges, eyes, nose, mouth, facial expressions, and textures. A shallow neural network (having only one or two hidden layers) cannot learn these complex features effectively, resulting in poor recognition accuracy.

To improve the performance of the facial recognition system, the network architecture should be modified by increasing the network depth and selecting suitable activation functions.

Problems with a Shallow Neural Network

A shallow neural network has limited learning ability because:

- It contains very few hidden layers.
- It learns only simple features such as edges and lines.
- It cannot capture complex facial patterns.
- It has low feature extraction capability.
- It gives poor classification accuracy.
- It may underfit the training data.
- It performs poorly when face images have different lighting, pose, or facial expressions.

Recommended Architectural Changes

1. Increase the Network Depth

The first improvement is to use a deeper neural network by adding more hidden layers.

Why Increase Depth?

A deep neural network learns features in multiple stages.

- **First Hidden Layers**
 - Learn simple features
 - Detect edges and corners
- **Middle Hidden Layers**
 - Learn facial parts
 - Detect eyes, nose, ears, lips
- **Deeper Hidden Layers**
 - Learn complete face structure
 - Recognize facial identity and expressions

As the depth increases, the model becomes capable of learning more meaningful and complex representations of facial images.

Benefits of Increasing Network Depth

- Learns hierarchical features.
- Captures complex facial structures.
- Improves feature extraction.
- Increases recognition accuracy.
- Better handles pose and lighting variations.
- Improves generalization on unseen faces.

2. Choose Proper Activation Functions

Activation functions introduce non-linearity into the neural network, allowing it to learn complex relationships.

Use ReLU in Hidden Layers

The recommended activation function is ReLU (Rectified Linear Unit).

Formula

$$\text{ReLU}(x) = \max(0, x)$$

- Faster training.
- Simple to compute.
- Reduces vanishing gradient problem.
- Activates only important neurons.
- Improves convergence.
- Works well in deep neural networks.

Use Softmax in the Output Layer

Since facial recognition is a multi-class classification problem, the output layer should use Softmax Activation.

Purpose of Softmax

- Converts output values into probabilities.
- Predicts the most likely person's identity.
- Produces probability values between 0 and 1.
- The class with the highest probability is selected.

3. Increase Network Width (Optional Improvement)

Network width means increasing the number of neurons in each hidden layer.

Advantages

- Learns more features at each layer.
- Improves representation of facial images.
- Handles larger datasets effectively.

However, increasing width alone is usually less effective than increasing depth.

4. Additional Improvements

To further improve facial recognition:

Batch Normalization

- Stabilizes training.
- Speeds up learning.
- Reduces internal covariate shift.

Dropout

- Randomly disables neurons during training.
- Prevents overfitting.
- Improves model generalization.

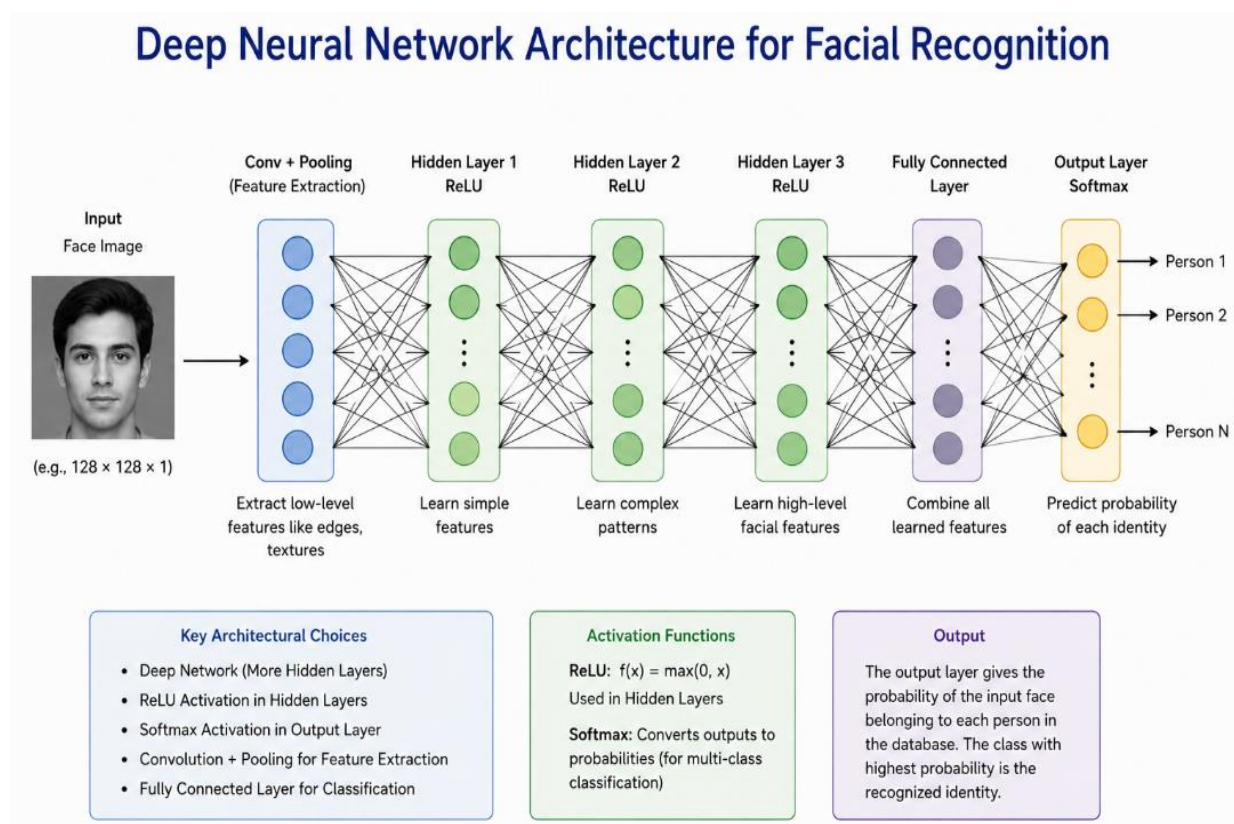
Data Augmentation

Generate more training images by:

- Rotating images
- Flipping images
- Cropping images
- Changing brightness
- Zooming

This increases the diversity of training data.

Deep Learning Architecture:



Advantages of Deep Networks with ReLU:

- Learns complex facial features.
- Higher recognition accuracy.
- Faster training.
- Handles different lighting conditions.
- Better performance on large datasets.
- Reduces vanishing gradient problem.
- Improves feature learning.
- Better generalization on new face images.

Conclusion

A shallow neural network is not sufficient for facial recognition because it cannot learn

complex facial patterns. Increasing the network depth allows the model to learn hierarchical features, while using ReLU in hidden layers improves training efficiency and reduces the vanishing gradient problem. Softmax in the output layer enables accurate face classification. Together, these architectural improvements significantly enhance the performance, accuracy, and robustness of the facial recognition system.

4.Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer:

Introduction

Streaming platforms such as Netflix or Amazon Prime collect a large amount of user viewing history, including movies watched, watch time, and viewing frequency. Since this data is unlabeled (there are no categories like "liked" or "disliked"), an RBM (Restricted Boltzmann Machine) can be used to learn hidden user preferences automatically.

An RBM is an unsupervised deep learning model that learns relationships between users and movies. It discovers hidden patterns in viewing history and predicts movies that a user is likely to enjoy.

Why Use RBMs?

RBMs are suitable because they:

- Work with unlabeled data.
- Learn hidden relationships between users and movies.
- Automatically discover user preferences.
- Predict movies that users have not watched yet.
- Improve recommendation accuracy.
- Handle large datasets efficiently.

Architecture of an RBM

An RBM consists of **two layers**:

1. Visible Layer (Input Layer)

The visible layer contains observed data.

For movie recommendation, it represents:

- Movies watched
- User ratings (if available)
- Viewing history
- Watch frequency

Each neuron represents one movie.

Example:

Movie A = 1 (Watched) Movie B = 0 (Not Watched)

Movie C = 1 (Watched)

2. Hidden Layer

The hidden layer learns hidden features automatically.

These features may represent:

- Action movie preference
- Comedy preference
- Romance interest
- Favorite actors
- Preferred language
- Watching habits
- Genre preference

These hidden features are not manually defined; the RBM learns them during training.

Working Process

Step 1: Collect Viewing History

The streaming platform collects data such as:

- Movies watched
- Number of times watched
- Viewing duration
- Recently watched movies

This information forms the user-movie interaction matrix.

Step 2: Prepare Input Data

The viewing history is converted into a matrix.

Example:

User	Movie A	Movie B	Movie C	Movie D
User 1	1	0	1	0
User 2	0	1	1	1
User 3	1	1	0	0

- **1 = Watched**
- **0 = Not Watched**

This matrix becomes the input to the RBM.

Step 3: Unsupervised Training

The RBM is trained using Contrastive Divergence (CD).

During training:

- The visible layer sends information to the hidden layer.
- The hidden layer learns hidden viewing preferences.
- The decoder reconstructs the original viewing pattern.
- The reconstruction error is minimized.

Since the data is unlabeled, the RBM learns patterns automatically.

Step 4: Learn Hidden Features

After training, the hidden neurons capture important customer interests such as:

- Favorite movie genres
- Preferred actors
- Watching time
- Similar movie choices
- Viewing habits

These learned features represent each user's interests.

Step 5: Predict Missing Movies

The trained RBM estimates the probability that a user will like movies they have not watched.

Example:

Movie	Predicted Score
Movie X	0.95
Movie Y	0.87
Movie Z	0.20

Higher score = Higher chance the user will enjoy the movie.

Step 6: Generate Recommendations

The system recommends the Top-N movies with the highest predicted scores.

Example:

- Movie X
- Movie Y
- Movie P
- Movie Q
- Movie R

These recommendations are personalized for each user.

Advantages of RBMs

- Works without labeled data.
- Learns hidden user preferences automatically.
- Captures complex user–movie relationships.
- Produces personalized recommendations.
- Handles large and sparse datasets.
- Improves recommendation accuracy.
- Reduces manual feature engineering.
- Scalable for millions of users and movies.

Applications of RBMs

- Movie recommendation systems
- Music recommendation

- Product recommendation
- E-commerce platforms
- Online video streaming
- Personalized content recommendation

Conclusion

Restricted Boltzmann Machines (RBMs) are powerful unsupervised deep learning models used in recommendation systems. They learn hidden patterns from users' viewing history without requiring labeled data. By discovering user preferences and predicting unseen movies, RBMs generate accurate and personalized movie recommendations, improving user satisfaction and engagement.

5.Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer:

Introduction:

In modern manufacturing industries, machines continuously generate sensor readings such as temperature, pressure, vibration, speed, voltage, and sound every second. These sensors produce a huge amount of data, making manual inspection difficult. A deep learning model based on Representation Learning and Neural Networks can automatically learn important patterns from the sensor data and accurately identify defective products in real time.

Recommended Deep Learning Solution:

The best solution is to use an Autoencoder, which is a deep learning model used for Representation Learning.

An Autoencoder automatically learns the hidden features of normal sensor readings and identifies defective products by measuring the reconstruction error. Products with high reconstruction error are considered defective.

Why Representation Learning?

Representation Learning is suitable because it:

- Automatically learns useful features from raw sensor data.
- Removes unnecessary and noisy information.
- Reduces manual feature engineering.
- Improves defect detection accuracy.
- Works well with large amounts of sensor data.
- Learns hidden relationships between different sensor values.

Proposed Deep Learning Architecture

The proposed system consists of the following stages:

1. Sensor Data Collection

Sensors installed on manufacturing machines continuously collect:

- Temperature
- Pressure
- Vibration
- Speed
- Voltage
- Current
- Sound
- Humidity

Thousands of readings are collected every second.

2. Data Preprocessing

Before training the neural network, the sensor data is prepared by:

- Removing missing values.
- Removing noise.
- Normalizing sensor values.
- Converting data into fixed-length time windows.

This improves learning accuracy.

3. Representation Learning (Encoder)

The Encoder learns important features from the sensor readings.

Instead of storing hundreds of sensor values, it compresses them into a small set of meaningful features.

Example:

500 Sensor Features

↓

40 Important Features

These compressed features represent the normal operating condition of the machine.

4. Latent Representation

The compressed data is called the latent representation.

It stores:

- Machine operating behavior
- Sensor relationships
- Hidden fault patterns
- Product quality information

This layer contains only the most useful information.

5. Decoder

The Decoder reconstructs the original sensor readings from the compressed features.

The reconstructed output is compared with the original sensor data.

The difference is called the Reconstruction Error.

6. Defect Detection

For every new product:

- Sensor data is passed through the trained Autoencoder.
- Reconstruction error is calculated.

If

- Low Reconstruction Error → Normal Product
- High Reconstruction Error → Defective Product

This allows automatic defect detection.

7. Real-Time Monitoring

The trained neural network is connected directly to the production line.

As each product passes through the machine:

- Sensor readings are collected.
- Deep learning model analyzes the data.
- Defective products are immediately detected.
- Faulty products are removed automatically.

Role of Neural Networks

The neural network performs the following tasks:

- Learns complex sensor patterns.
- Extracts meaningful features automatically.
- Recognizes hidden defects.
- Improves prediction accuracy.
- Reduces manual inspection.

Advantages

- Detects defective products automatically.
- Works in real time.
- Reduces human inspection.
- Learns features automatically.
- Handles large sensor datasets.
- Improves production quality.
- Reduces manufacturing cost.
- Detects defects at an early stage.
- Improves customer satisfaction.
- Scalable for smart factories.

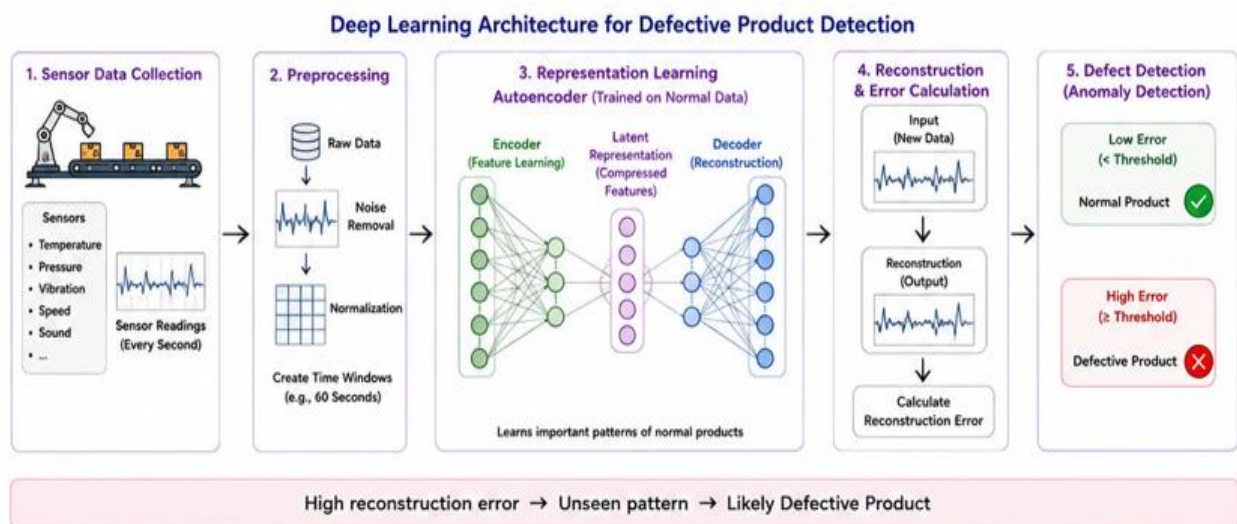
Applications

- Manufacturing industries
- Smart factories
- Predictive maintenance

- Machine health monitoring
- Quality control systems
- Industrial IoT
- Automotive manufacturing
- Electronics production

Conclusion

Representation Learning combined with Neural Networks provides an efficient solution for detecting defective products from continuous sensor readings. An Autoencoder learns the hidden patterns of normal machine behavior and identifies abnormal products using reconstruction error. This deep learning approach improves accuracy, enables real-time quality monitoring, reduces production losses, and increases overall manufacturing efficiency.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially	No meaningful relationships established

		between concepts		incorrect	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand